

Feature Engineering Report

Dataset Overview

The objective is to predict whether a trip is a cutoff (``is_cutoff``) or not using various features derived from trip time, route, delays, and center pair combinations.

Exploratory Data Analysis (EDA)

Performed basic inspection using `shape()`, `.info()`, `.describe()`, and unique value checks. This helps understand missing data, data types, and column distributions for preprocessing.

Feature Selection

- Correlation matrix and VIF were used to identify multicollinearity.
- Features with high correlation (> 0.9) and $VIF > 10$ were dropped to improve model generalization.
- Mutual Information and Random Forest were used to identify important features for classification.

Feature Creation

- Extracted datetime parts (hour, weekday, etc.) for time-based insights.
- Created delay-related ratios, time differences, and route-level aggregations.
- Created domain-specific indicators like ``is_heavy_delay`` and ``delay_category`` using thresholds.

Feature Transformation

- Applied `MinMaxScaler`, `StandardScaler`, `Log Transform`, `QuantileTransformer`, and `PolynomialFeatures`.
- Each transformation normalizes or enhances features for better learning behavior in ML models.

Feature Scaling

- Applied multiple scalers (Standard, Robust, MaxAbs, MinMax) to numeric features.
- Combined scaled results for experimentation with different ML algorithms.

Feature Reduction

- PCA reduced features while retaining 95% of the variance.
- Univariate methods (ANOVA F-test, Chi-square) and embedded methods (LassoCV, RFE) selected top features.

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- This enhances model performance and reduces overfitting risk.

Saved Outputs

- feature_engineered_dataset.csv: Post custom feature creation
- transformed_features_dataset.csv: After all transformations
- scaled_features_dataset.csv: Various scalings
- pca_reduced_dataset.csv: Dimensionality reduced with PCA

Summary

This pipeline includes domain-specific and statistical feature engineering, scaling, and reduction techniques. It enables training high-performance machine learning models by improving data quality, relevance, and interpretability.